

Calc Unit 4 CRAM Session

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Warm Up!

$x^2 - 9x + 19$ intersects its derivative at $x = a, b$. If $a < b$, find $a + 9b$.

4.2: Extreme Values

Key Terms:

Absolute maximum: greatest value of $f(x)$ in the given interval I .

Abs minimum: lowest value of $f(x)$

Local maximum: $f(x)$ value is higher than the values around it

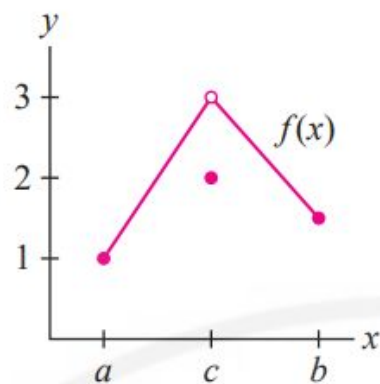
Local minimum: $f(x)$ value is lower than those around it

Critical point: c (in the domain of $f(x)$) is a critical point if $f'(c) = 0$ or doesn't exist

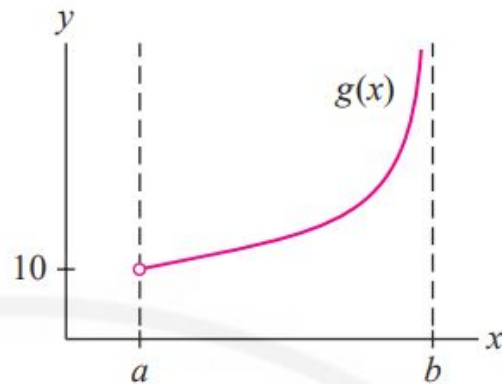
Fermat's Theorem: If $f(c)$ is a local max or min, c is a critical point of $f(x)$

Extreme Values Theorem: If $f(x)$ is continuous on $[a,b]$, and $f(c)$ is a min/max, c is either an endpoint (a or b) or a critical point

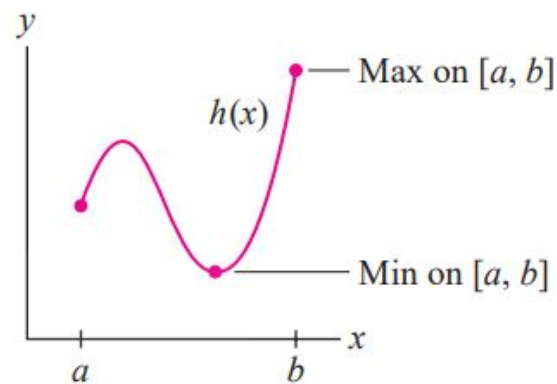
Rolle's Theorem: If $f(x)$ is continuous on $[a,b]$ and differentiable on (a,b) , and $f(a)=f(b)$, then there is a value c between a and b such as $f'(c)=0$



(A) Discontinuous function with no max on $[a, b]$.



(B) Continuous function with no min or max on the open interval (a, b) .



(C) Every continuous function on a closed interval $[a, b]$ has both a min and a max on $[a, b]$.

FIGURE 2

759. Let

$$f(x) = x(x+1)(x+2)(x+3).$$

Show that the equation

$$f'(x) = 0$$

has three real roots.

760. The equation

$$e^x = 1 + x$$

obviously has a root $x=0$. Show that this equation cannot have any other real root.

4.3: MVT and Monotonicity

Mean Value Theorem: If $f(x)$ is continuous on $[a,b]$ and differentiable on (a,b) :

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Monotonicity:

If $f'(x) > 0$ for $x \in (a, b)$, then f is increasing on (a, b) .

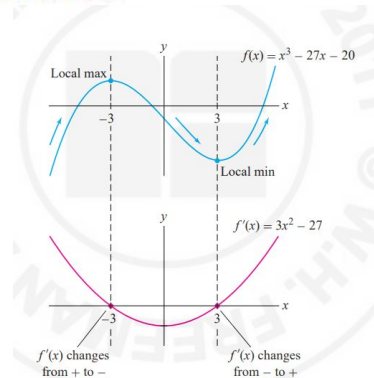
If $f'(x) < 0$ for $x \in (a, b)$, then f is decreasing on (a, b) .

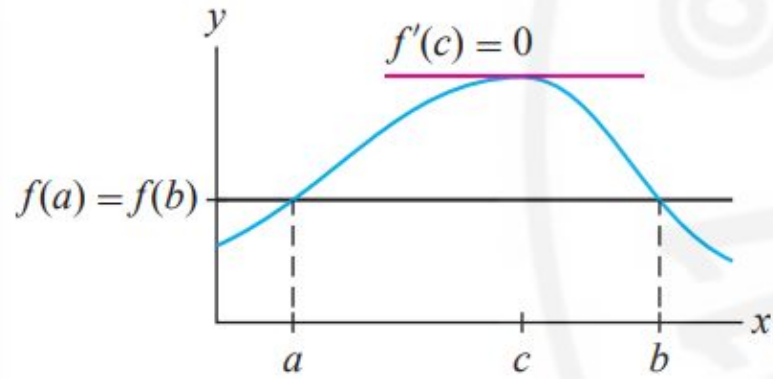
First Derivative Test:

Assume c is a critical point

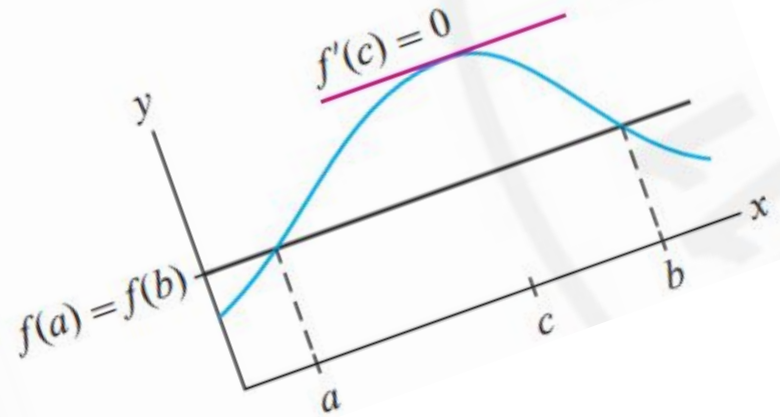
If $f'(x)$ changes from positive to negative, $f(c)$ is a local max

If $f'(x)$ changes from negative to positive, $f(c)$ is a local min





Rolle's Theorem



MVT

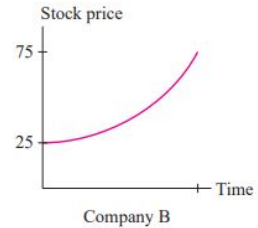
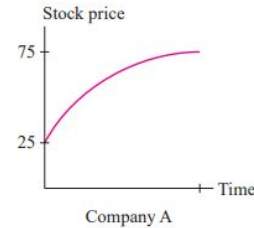
4.4 Shape of the Graph

Concavity: Function is concave up on (a,b) if $f'(x)$ is increasing on (a,b) ($f''(x)$ is positive), function is concave down on (a,b) if $f'(x)$ is decreasing on (a,b)

Test for Inflection: If $f''(c)$ is 0, AND $f''(x)$ changes sign @ c , $f(x)$ has a POINT OF INFLECTION (where the function changes concavity) at c .

Second Derivative Test: c is a critical point

- If $f''(c) > 0$, $f(c)$ is a local minimum
- If $f''(c) < 0$, $f(c)$ is a local maximum



4.4: The Shape of Graph

x	0	$(0, 1)$	1	$(1, e)$	$e \approx 2.72$	$(e, e^{\frac{3}{2}})$	$e^{\frac{3}{2}} \approx 4.49$	$(e^{\frac{3}{2}}, +\infty)$
y	$-\infty$	$-$	0	$+$	$\frac{1}{e} \approx 0.37$	$+$	$\frac{3}{2\sqrt{e^3}} \approx 0.33$	$+$
y'	nonexist.	$-$	$+$	$+$	0	$-$	$-$	$-$
y''	nonexist	$-$	$-$	$-$	$-$	$-$	0	$+$
Conclusions	Boundary point of domain or del. of function Vertical asymptote	Function increases; graph is concave down	Function has a local max; graph is concave down	Function increases; graph is concave down	Max point of function	Function decreases; graph is concave down	Point of inflection	Function decreases; graph is concave up

4.5: L'Hospital Rule

L'Hopital's Rule:

- $f(x)$ and $g(x)$ are differentiable on open interval containing a
- $g'(x)$ is NOT equal to 0 (except maybe at a)
- $f(x)/g(x)$ MUST produce an indeterminate form ($0/0$ or infinity/infinity)

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$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

2°. **Other indeterminate forms.** To evaluate an indeterminate form like $0 \cdot \infty$, transform the appropriate product $f_1(x) \cdot f_2(x)$, where $\lim_{x \rightarrow a} f_1(x) = 0$ and

$\lim_{x \rightarrow a} f_2(x) = \infty$, into the quotient $\frac{f_1(x)}{\frac{1}{f_2(x)}}$ (the form $\frac{0}{0}$ (or $\frac{\frac{f_2(x)}{1}}{f_1(x)}$ (the form $\frac{\infty}{\infty}$)).

In the case of the indeterminate form $\infty - \infty$, one should transform the appropriate difference $f_1(x) - f_2(x)$ into the product $f_1(x) \left[1 - \frac{f_2(x)}{f_1(x)} \right]$ and first evaluate the indeterminate form $\frac{f_2(x)}{f_1(x)}$; if $\lim_{x \rightarrow a} \frac{f_2(x)}{f_1(x)} = 1$, then we reduce the expression to the form

$$\frac{1 - \frac{f_2(x)}{f_1(x)}}{\frac{1}{f_1(x)}} \quad (\text{the form } \frac{0}{0}).$$

The indeterminate forms 1^∞ , 0^0 , ∞^0 are evaluated by first taking logarithms and then finding the limit of the logarithm of the power $[f_1(x)]^{f_2(x)}$ (which requires evaluating a form like $0 \cdot \infty$).

4.6 Graph Sketching and Asymptotes

DISARMIT

- **Domain/range, Intercepts, Symmetry, Asymptotes, Relative/Abs Extrema, Monotonicity/Concavity, Limits at Infinity, Table**
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4.7: Applied Optimization

Basic Steps

- i) Find the critical points of $f(x)$ in $[a,b]$
- ii) Evaluate $f(x)$ at the critical points and the endpoints a and b
- iii) The largest/smallest values are the extreme values of $f(x)$.

Tips:

- Find the domain beforehand so you don't accidentally find solutions outside the domain (ex. A side length cannot be negative)
- The hardest part of this is finding the equation relating the variables. Make sure to brush up on geometry and trig formulas.

4.8 Newton's Method

Purpose: To find approximations of zero's.

Newton's Method To approximate a root of $f(x) = 0$:

Step 1. Choose initial guess x_0 (close to the desired root if possible).

Step 2. Generate successive approximations $x_1, x_2, \dots$, where

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

- This can be using to approximate things such as sqrts.
- When finding $\text{sqrt}(5)$, use the function: x^2-5 and initial guess 2.

4.1 Linearization

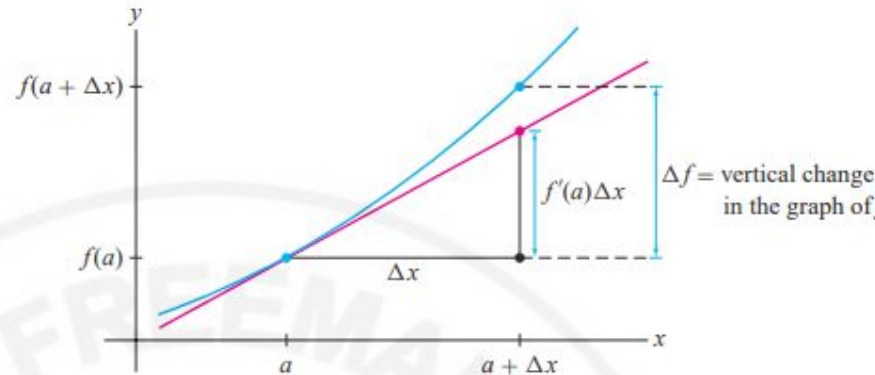
Linear Approximation of Δf If f is differentiable at $x = a$ and Δx is small, then

$$\Delta f \approx f'(a)\Delta x$$

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where $\Delta f = f(a + \Delta x) - f(a)$.

- This essentially creates a tangent line centered around $f(a)$, which is useful for approximating nearby values.



Approximate $\sin(10^\circ)$.

The value of a stock after x years is given by x^x dollars/unit (in thousands). After roughly how many months should one invest in the stock?

Find the value of k for which $f(x) = 4kx^3 - k^2x^2$ has an inflection point at $x = 1$.

What is the maximum volume of a cone inscribed in a sphere of radius r ?

Which cylinder has the smallest surface area, given a particular volume?

A point X touching one side of a river of uniform width 140 meters is 500 meters away from a point Y on the opposite side of the river, which also touches the river. If the price to travel along the river is \$ 0.67 per meter and the price to travel across the surrounding land is \$ 0.41 per meter, what is the cost of the cheapest trip from X to Y?

$$\lim_{x \rightarrow a} \frac{x - a}{\ln x - \ln a}$$

$$\lim_{x \rightarrow \infty} (\sinh x - \cosh x)$$

$$\lim_{x \rightarrow \infty} \frac{\tanh x}{\tan^{-1} x}$$

$$\lim_{x \rightarrow 2} \frac{e^{x^2} - e^4}{x - 2}$$

$$\lim_{x \rightarrow 0} \left(\frac{e^x}{e^x - 1} - \frac{1}{x} \right)$$

$$\lim_{x \rightarrow \infty} \frac{\sqrt[3]{x}}{\ln x}$$

Example 1. Compute

$$\lim_{x \rightarrow 0} \frac{\ln x}{\cot x} \text{ (form } \frac{\infty}{\infty}\text{)}.$$

Solution. Applying the L'Hospital rule we have

$$\lim_{x \rightarrow 0} \frac{\ln x}{\cot x} = \lim_{x \rightarrow 0} \frac{(\ln x)'}{(\cot x)'} = - \lim_{x \rightarrow 0} \frac{\sin^2 x}{x}.$$

We get the indeterminate form $\frac{0}{0}$; however, we do not need to use the L'Hospital rule, since

$$\lim_{x \rightarrow 0} \frac{\sin^2 x}{x} = \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \sin x = 1 \cdot 0 = 0.$$

We thus finally get

$$\lim_{x \rightarrow 0} \frac{\ln x}{\cot x} = 0.$$

Example 2. Compute

$$\lim_{x \rightarrow 0} \left(\frac{1}{\sin^2 x} - \frac{1}{x^2} \right) \text{ (form } \infty - \infty \text{)}.$$

Example 3. Compute

$$\lim_{x \rightarrow 0} (\cos 2x)^{\frac{3}{x^2}} \text{ (form } 1^\infty \text{)}$$

786. $\lim_{x \rightarrow 0} \frac{\ln (\sin mx)}{\ln \sin x}$

$$\bar{y} = 2x^3 + 3x^2 - 12x + 5.$$

DISARM

29. $\lim_{x \rightarrow -1} \frac{\sqrt{x+10} + 3x^{1/3}}{4x^2 + 3x - 1}$

30. $\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$

4. Find the positive, real value of k where $e^{kx} = 3\sqrt{x}$ has exactly 1 solution.

$$\sum_{n=0}^{\infty} \left(\sqrt{n^2 + 3n + 2} - \sqrt{n^2 + n} - 1 \right) .$$

$$\lim_{x \rightarrow \infty} x^{1/(\ln x)}$$

984. $y = \arctan (\ln x).$

985. $y = \arcsin \ln (x^2 + 1)$

986. $y = x^x.$

987. $y = x^{\frac{1}{x}}.$